

■ Agentic Protocol Engineering

The Wiring Harness of Trust

■ The Production Reality Check

Clean whiteboard diagrams hide dangerous gaps in agentic systems where trust leaks through the seams.

- What Fails in Production:
- Timestamps vanish between retrieval and reasoning
 - Provenance gets stripped from critical data
 - Memory drifts without structured handoffs
 - Actions fire without governance checks
 - Tools execute with unchecked privileges

■ The Protocol Solution: Trust That Travels

Protocol	What It Does	Why It Matters
■ MCP Model Context	Versioned, structured context slices	Models get predictable, validated inputs
■ ACP Agent Communication	Traceable message envelopes	Every conversation leaves an audit trail
■ A2A Agent-to-Agent	Task coordination & state sync	Autonomous agents work together safely
■ ANP Agent Network	Cross-org identity & policy	Trust extends beyond enterprise walls

■ Maturity Ladder: From Fragile to Bulletproof

■ L0: Ad Hoc Chaos → ■ L1: Sandbox Safety → ■ L2: Context Stability → ■ L3: Structured Communication → ■ L4: Federated Trust → ■ L5: Protocol Mesh

■ Protocol Envelope Example

```
{ "protocol": "ACP_Message", "metadata": { "id": "msg_1774875876", "session":  
"med_review_001", "timestamp": "2026-03-30T15:04:36Z" }, "payload": { "tool":  
"generate_report", "execution_class": "restricted", "provenance": { "origin":  
"verified_medical_db", "trust_score": 0.95 }, "governance": {  
"requires_approval": true, "approver_role": "Chief_Medical_Officer" } } }
```

■ *Key Takeaway*

APIs connect components. Protocols connect trust.

"The diagram is just a map. The protocols are the wiring harness that keeps the whole thing intact when it moves."

— Austin & Neter, Chapter 8